MATH MA'AM ACADEMY

Where Every Subject Clicks.

STD 5 MATHEMATICS – CHAPTER 1: NUMBER SENSE

Learning objectives: Numerals and Number Names • Place Value and Face Value • Roman Numbers • Estimation • Formation of Numbers • Ascending and Descending Order

1.1 NUMERALS AND NUMBER NAMES

Numeral: A number written using digits. **Number name:** A number written in words. The digits 0–9 are used to write numbers.

Indian System of Numeration

In the Indian system, digits are grouped into periods: Ones, Thousands, Lakhs and Crores. Commas are placed after the ones period (three digits from the right), and then after every two digits.

Period	Places included	Example grouping
Crores	Ten crores, Crores	15
Lakhs	Ten lakhs, Lakhs	40
Thousands	Ten thousands, Thousands	39
Ones	Hundreds, Tens, Ones	846

Example: 154039846 → **15,40,39,846**
Number name: Fifteen crore forty lakh thirty-nine thousand eight hundred forty-six.

International System of Numeration

The International system groups digits in periods of three from the right: Ones, Thousands and Millions. Commas are inserted after every three digits.

Period	Places included	Example grouping
Millions	Hundred millions, Ten millions, Millions	154
Thousands	Hundred thousands, Ten thousands, Thousands	039
Ones	Hundreds, Tens, Ones	846

Example: 154039846 → **154,039,846**
Number name: One hundred fifty-four million thirty-nine thousand eight hundred forty-six.

Remember: The same number is written differently in the two systems: Indian – 15,40,39,846; International – 154,039,846.

1.2 PLACE VALUE AND FACE VALUE

Face value: The digit itself, wherever it is placed.
Place value: The value of a digit according to its position in the number.

Example: In 7,20,53,419:

Digit	Position	Place value	Face value
7	Crores	7,00,00,000	7
2	Ten lakhs	20,00,000	2
0	Lakhs	0	0
5	Ten thousands	50,000	5
3	Thousands	3,000	3
4	Hundreds	400	4
1	Tens	10	1
9	Ones	9	9

Expanded form: Write a number as the sum of the place values of its non-zero digits.
Short form: Write the expanded form as one numeral.

Example: 24,680,590 = 20,000,000 + 4,000,000 + 600,000 + 80,000 + 500 + 90.

Useful conversions:
10 ones = 1 ten • 10 tens = 1 hundred • 10 hundreds = 1 thousand
10 thousands = 1 ten thousand • 10 lakhs = 1 million • 1 crore = 10 million • 10 crores = 100 million.

Predecessor: The number just before a given number = number – 1.
Successor: The number just after a given number = number + 1.

1.3 ROMAN NUMERALS

Roman numerals use letters to represent numbers. The main symbols are:

Roman symbol	I	V	X	L	C	D	M
Value	1	5	10	50	100	500	1000

Rules for writing Roman numerals

• **Addition:** When a smaller or equal value symbol is written to the right of a larger symbol, add the values. Example: VI = 5 + 1 = 6.

• **Subtraction:** When a smaller value symbol is written to the left of a larger symbol, subtract it. Example: IV = 5 – 1 = 4; IX = 10 – 1 = 9.

• **Repetition:** I, X, C and M can be repeated, generally up to three times in succession. V, L and D are not repeated.

• **Zero:** There is no Roman numeral symbol for zero.

• **Common subtraction pairs:** IV = 4, IX = 9, XL = 40, XC = 90, CD = 400, CM = 900.

Examples: 49 = XLIX; 94 = XCIV; 444 = CDXLIV; 1,984 = MCMLXXXIV; 2,769 = MMDCCLXIX.

Check carefully: A symbol should not be repeated more than the allowed number of times. For example, 9 is IX, not VIIII.

1.4 ESTIMATION AND ROUNDING OFF

Rounding gives an approximate value that is easier to use. To round a number to a particular place:

• Look at the digit immediately to the right of the place to which you are rounding.

• If that digit is 0, 1, 2, 3 or 4, keep the rounding digit unchanged and replace digits to its right with zeros.

• If that digit is 5, 6, 7, 8 or 9, increase the rounding digit by 1 and replace digits to its right with zeros.

Worked examples

Number	Nearest ten	Nearest hundred	Nearest thousand
4,256,892	4,256,890	4,256,900	4,257,000
68,749	68,750	68,700	69,000
876,500	876,500	876,500	877,000

Finding possible numbers: Numbers that round to 35,000 to the nearest thousand range from 34,500 to 35,499. Therefore, the smallest possible whole number is 34,500.

1.5 FORMATION OF NUMBERS

Numbers can be formed using given digits, either with repetition or without repetition.

Without repetition

• **Greatest number:** Arrange all digits in descending order (largest to smallest).

• **Smallest number:** Put the smallest non-zero digit first, then arrange the remaining digits in ascending order. Zero cannot be placed at the beginning.

Example: Using 5, 2, 4, 0, 1, 3 and 6 exactly once:
Greatest = 6543210
Smallest = 1023456

With repetition

• If repetition is allowed, a digit may be used more than once, subject to the question's conditions.

• For the greatest number, repeat the greatest available digit in every place if no other restriction applies.

• For the smallest number, the first digit must be the smallest non-zero digit; fill the remaining places with the smallest permitted digit.

Special conditions

• **Greatest even number:** The last digit must be even. Arrange the other digits to make the number as large as possible.

• **Smallest even number:** The last digit must be even; choose the smallest possible leading digit and arrange the remaining digits carefully.

• **Odd number:** The last digit must be 1, 3, 5, 7 or 9.

Common mistake: A number cannot begin with zero. For example, the smallest 5-digit number using 0, 1, 2, 5 and 9 is 10,259, not 01,259.

1.6 ASCENDING AND DESCENDING ORDER

Ascending order: Arrange numbers from smallest to greatest.
Descending order: Arrange numbers from greatest to smallest.

How to compare large numbers

• First compare the number of digits. A number with more digits is greater.

• If the numbers have the same number of digits, compare digits from the leftmost place.

• At the first place where the digits differ, the number with the greater digit is the greater number.

Example: Given 4,567,251; 5,210,768; 3,459,612:
Ascending: 3,459,612; 4,567,251; 5,210,768
Descending: 5,210,768; 4,567,251; 3,459,612

QUICK REVISION – FORMULAE AND FACTS

Concept	Remember
Place value	Digit × value of its place
Face value	The digit itself
Predecessor	Given number – 1
Successor	Given number + 1
Indian commas	First group of 3 digits, then groups of 2
International commas	Groups of 3 digits from the right
1 crore	10 million
10 lakhs	1 million
Rounding 0–4	Keep the rounding digit unchanged
Rounding 5–9	Increase the rounding digit by 1

Practice tip: Before an IMO examination, practise reading large numerals in both systems, identifying place values, converting Roman numerals, rounding numbers, and forming the greatest and smallest numbers under different conditions.

ALL THE BEST